

Contact

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www.linkedin.com/in/valentine-sinev (LinkedIn)
github.com/valiksinev (Personal)
hh.ru/resume/db5c2fdbff0bc387bf0039ed1f77574f5 (Portfolio)

Top Skills

Rust (Programming Language)
Solana
EVM

Patents

GLASS CLEANLINESS SENSOR

Valentine Sinev

Rust Developer
Russia

Summary

I'm a Rust developer working at the intersection of Ethereum and Solana. At Rome Protocol, I build Rome EVM — an on-chain EVM execution engine written in Rust that runs Ethereum smart contracts on Solana. Any contract that works on Ethereum works here, with the same tools and the same semantics. The work touches every layer of the Solana stack: account model constraints, parallel execution patterns, ALT, CPI, secure asset vault design. I enjoy working where the complexity is real.

Experience

Rome Protocol
Core EVM Engineer
April 2024 - Present (2 years 1 month)

Designed and built Rome EVM from scratch — an Ethereum-equivalent execution layer running natively on Solana. Sole architect of the on-chain program, off-chain emulator, and transaction sender.

The central engineering challenge: Solana's account model, transaction size limits, and execution constraints are fundamentally incompatible with Ethereum semantics. Built the systems to bridge that gap without compromising EVM compatibility:

- Iterative transaction execution — splits long EVM runs across multiple Solana transactions, serializing full VM state between steps
- Gas model based on Solana rules,
- Address Lookup Table integration for transactions requiring large account sets
- Using a custom heap allocator,

- SBF emulator for off-chain simulation of on-chain logic,
- Solidity interface for cross-program invocation — contracts can call native Solana programs directly from EVM bytecode

Any Ethereum contract, tool, or wallet (MetaMask, ethers.js, Hardhat, Foundry) works without modification.

Neon Labs

Rust Developer

January 2021 - August 2023 (2 years 8 months)

Russia

Participation in the development of the Neon-EVM product for the Solana blockchain.

Development an application for transaction emulation inside a Solana BPF virtual machine.

Participation in the development of the tracer-api service. This service implements the EIP-1898 and Geth interfaces, collects traces, allows you to execute transactions based on the historical state.

Implementation of EIP-198 as the syscall of the Solana node (<https://github.com/solana-labs/solana/pull/28503>).

oneFactor

C++ Developer

April 2019 - January 2021 (1 year 10 months)

Moscow, Russia

Design and development from scratch of a service based on Intel SGX technology. Implementation of remote attestation, local attestation of the enclave. Development and implementation of a cryptographic scheme.

Krypton Moscow

C++ Developer

May 2018 - March 2019 (11 months)

Moscow, Russia

Development of an application for signing transactions and storing keys

<https://github.com/arrayio/array-io-keychain>

Implementation of the backup/restore scheme for keys (BIP-32, BIP-39, BIP-44). Implementation of transaction signatures Ethereum, Bitcoin, Bitshares, ArrayIO.

Bell Integrator

C++ Developer

July 2017 - April 2018 (10 months)

Moscow, Russia

participation in the development of the project for the National Settlement Depository

Sinkross

C Developer

July 2009 - May 2017 (7 years 11 months)

Saratov, Saratov region, Russia

Development of software for industrial automation, sensor firmware.

Development of signal processing algorithms.

Education

Saratov State Technical University

Magistracy, Production automation · (1997 - 2003)